

Practical 8

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

 /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003

```
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, faults = 0, hits = 0;
    int flag1, flag2;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag1 = flag2 = 0;

        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                flag1 = flag2 = 1;
                break;
            }
        }

        if(flag1 == 0) {
            for(j = 0; j < framesCount; j++) {
                if(frames[j] == -1) {
                    faults++;
                    frames[j] = pages[i];
                    flag2 = 1;
                    break;
                }
            }
        }

        if(flag2 == 0) {
            int least[10];
            for(j = 0; j < framesCount; j++) {
```

 Read

```
GNU nano 8.7
frames[j] = pages[i];
flag2 = 1;
break;
    }
}
}
if(flag2 == 0) {
    int least[10];
    for(j = 0; j < framesCount; j++) {
        least[j] = -1;
        for(k = i - 1; k >= 0; k--) {
            if(frames[j] == pages[k]) {
                least[j] = k;
                break;
            }
        }
    }
    pos = 0;
    for(j = 1; j < framesCount; j++)
        if(least[j] < least[pos])
            pos = j;
    frames[pos] = pages[i];
    faults++;
}

printf("Frame status: ");
for(j = 0; j < framesCount; j++)
    printf("%d ", frames[j]);
printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}
```

Output:

 /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003

```
Chanchal@DESKTOP-GARLJAI MSYS ~
# cd /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003/

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# nano p8.c

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# gcc p8.c -o p8

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# ./p8
Enter number of pages: 1 2 3 4 1 2 5 1 2 3 4 5
Enter page reference string:
Frame status: 2 -1 -1

Total Page Hits = 0
Total Page Faults = 1
Hit Percentage = 0.00%
Fault Percentage = 100.00%

Chanchal@DESKTOP-GARLJAI MSYS /C/Users/Administrator/Desktop/OSY_CD25D003/OSY_CD25D003
# |
```